

Setting certain fields in NBT to invalid characters causes exceptions to be logged, potentially causing lag

MC-274047 Commands

Status	Resolved / Fixed
Affected	1.21, 1.21.1 Release Candidate 1, 1.21.1, 24w33a, 24w34a, 24w35a (+27 more)
Fixed in	25w07a
Reported	2024-07-01

STEPS TO REPRODUCE

1. Launch Minecraft and create a Creative world with default settings.
2. Run `/give @s minecraft:suspicious_sand[minecraft:block_entity_data={id:brushable_block,LootTable:"?"}]``.
3. Place the suspicious sand given by the command.
4. Run `/summon minecraft:falling_block ~ ~ ~ {BlockState:{"Name":"?"}}``.
5. Run `/summon minecraft:falling_block ~ ~ ~ {BlockState:{"Name":"suspicious_sand"},TileEntityData:{LootTable:"?"}}``.
6. Place a monster spawner below you.
7. Run `/data modify block ~ ~-1 ~ SpawnData.entity.inBlockState set value {Name:"?"}``.
8. Open the game's log files.
9. Observe whether stack trace exceptions are repeatedly logged.
10. Observe whether the exceptions cause noticeable FPS or TPS drops, including continued log spam while the Esc menu is open.

OBSERVED BEHAVIOUR

When an NBT field that is expected to contain a valid resource location is set to an invalid value, such as `"?"` or `"="`, the game accepts the data instead of rejecting it during command execution. The invalid value is later processed when the affected block, entity, or block entity is loaded or interacted with, causing an exception and stack trace to be written to the game log.

Examples include:

- Placing suspicious sand with `LootTable:"?"` causes an exception related to the brushable block's loot table.
- Spawning a falling block with `BlockState.Name:"?"` causes an exception and also displays an error in chat.
- A falling block with an invalid `TileEntityData.LootTable`` produces an exception while the entity is processed.
- A monster spawner containing an invalid `SpawnData.entity.inBlockState.Name`` repeatedly produces exceptions while the spawner attempts to use the invalid entity data. In singleplayer, this logging continues even while the game is paused on the Esc menu.

If the invalid data is processed repeatedly - for example, by a falling block summoned from a repeating command block or by a spawner - the same exception can be logged many times. This can rapidly increase the log file size and cause noticeable server tick-time/MSPT and client FPS performance degradation. In severe cases, the affected spawn egg fails to spawn the entity and an exception is logged when it is used.

EXPECTED BEHAVIOUR

When invalid characters are supplied in NBT fields such as `Name``, `LootTable``, or `RecipesUsed``, the game should handle the invalid value gracefully without repeatedly throwing or logging stack-trace exceptions. It should reject or safely ignore the malformed data, optionally provide a clear error message, and continue normal operation without causing excessive log growth, FPS drops, TPS degradation, or persistent errors from entities and spawners.

ENVIRONMENT

Minecraft Java Edition Versions: 1.21, 1.21.1 Release Candidate 1, 1.21.1, 24w33a, 24w34a, 24w35a, 24w36a, 24w37a, 24w38a, 24w39a, 24w40a, 1.21.2 Pre-Release 1, 1.21.2 Pre-Release 2, 1.21.2 Pre-Release 3, 1.21.2 Pre-Release 4, 1.21.2 Pre-Release 5, 1.21.2 Release Candidate 1, 1.21.2 Release Candidate 2, 1.21.2, 24w44a,

1.21.3, 24w45a, 1.21.4 Pre-Release 1, 1.21.4 Pre-Release 2, 1.21.4 Pre-Release 3, 1.21.4 Release Candidate 1, 1.21.4 Release Candidate 3, 1.21.4, 25w02a, 25w03a, 25w04a, 25w05a, 25w06a (includes snapshots)

ORIGINAL DESCRIPTION

The Bug

When setting certain NBT fields to invalid characters (such as ? and =), an exception is outputted in the log. A large amount of exceptions can cause lag to most systems.

Examples of some fields that cause errors when invalid characters are entered include RecipesUsed, LootTable, Name, and possibly more.

Steps to Reproduce (Direct)

Run the command specified for all situations below, then look in the game logs. Note if there is a stack trace exception or not.

Placing Suspicious Sand with invalid LootTable:

Place down the block this command gives you, and check the log.

```
/give @s minecraft:suspicious_sand[minecraft:block_entity_data={id:brushable_block,LootTable:"?"}]
```

FallingBlock entity with invalid Name:

This displays an error message in chat as well.

Running the command below in a repeating command block causes high MSPT lag.

```
/summon minecraft:falling_block ~ ~ ~ {BlockState:{Name:"?"}}
```

FallingBlock entity with invalid LootTable:

```
/summon minecraft:falling_block ~ ~ ~ {BlockState:{Name:"suspicious_sand"},TileEntityData:{LootTable:"?"}}
```

Steps to Reproduce (Spawner)

This may cause severe FPS lag.

Run the command below (attempting to right-click this spawn egg on the ground will also output an exception and not spawn anything):

```
/give @s minecraft:breeze_spawn_egg[minecraft:entity_data={id:arrow,inBlockState:{Name:"?"}}]
```

Place a monster spawner below you, and right-click the spawner with the spawn egg.

Run the command below while on top of the spawner:

```
/data modify block ~ ~-1 ~ SpawnData.entity.inBlockState set value {Name:"?"}
```

View the game logs and note if stack trace exceptions are spammed or not.

Observed Results

Exceptions are sometimes spammed to the game's log. In some cases, an FPS and TPS drop is noticeable. In the case of the spawner bug, errors continue to be spammed even when on the Esc menu in singleplayer.

The log file can also be huge due to these exceptions being spammed.

See the following text files for the exceptions experienced. They are all named respective to the thing which caused it.

- Placing Suspicious Sand with Invalid LootTable.txt
- FallingBlock entity with invalid Name.txt
- Fallin

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-274047>